

TOXICITY TEST WITH FRESH WATER ALGAE

Group 8

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ABSTRACT

A Water quality - Fresh water algal growth inhibition test was performed according to ISO standard 8692 to determine the inhibitory effects of 4-nitrophenol on the unicellular green algae *Raphidocelis subcapitata*.

Biomass development was quantified by fluorescence measurements of acetone-extracted chlorophyll at the start and end of the exposure, and growth rates were calculated for each concentration relative to the untreated control. The test was conducted over a 72-hour exposure period using five test concentrations and three replicates per concentration plus five control replicates.

The concentration range was selected around an expected effect concentration for 4-nitrophenol of 30 mg/L, and concentration–response relationships were analysed in R using a log-normal dose–response model. Validity criteria were evaluated by comparing control growth rate and pH stability with the protocol requirements. The data clearly showed a concentration-dependent inhibition, with concentrations approaching 120 mg/L resulting in complete inhibition of algal activity. Furthermore, the dataset retrieved from the fluorometer measurements allowed to define a complete dose-response curve capturing both low- and high-effect concentrations. Due to this the responses showed no sign of hormesis, in fact even at the lowest doses inhibition was occurring.

The estimated EC10, EC20 and EC50 values were 1.986, 3.588 and 11.12 mg/L, with 95% confidence intervals of [-0.6638, 4.636], [0.02986, 7.147], and [5.940, 16.31], respectively. Overall, the results indicate that the concentration range selected for the test was appropriate to obtain all the required EC values and fully characterize the toxic response of the algae. Together, these findings are essential for evaluating the ecological risk and the inhibitory effects of 4-nitrophenol on the test organism but also to define the regulatory thresholds that should be introduced for the compound.

INTRODUCTION

Raphidocelis subcapitata is a unicellular, planktonic freshwater green microalga commonly recommended by international guidelines as a standard test organism because of its rapid exponential growth and high sensitivity to a broad range of organic and inorganic toxicants [1]. Previously known as *Pseudokirchneriella subcapitata* and *Selenastrum capricornutum*, it serves as a representative primary producer in ecotoxicological studies and environmental risk assessments, including tests of metals, organic contaminants and complex wastewaters [2].

The compound 4-nitrophenol is characterized by moderate solubility (16 g/L at 25 °C), indicating that it is mobile and tends to remain in the aqueous phase. On the other hand, it has low volatility since both Henry's constant and vapor pressure are small (H <10^-5 atm-m3/mol and vapor pressure <0.01 mmHg). As a result, the compound is not expected to partition into the gas phase [3][4]. This makes 4-nitrophenol a safe compound to work with in the lab since inhalation is not expected to be a significant exposure pathway. Moreover, losses of the compound are not expected, as its very low volatility ensures that it remains within the vials.

In this procedure, algal cell biomass is quantified by measuring the fluorescence of acetone-extracted chlorophyll. The measured fluorescence serves as a highly sensitive, direct proxy for biomass density, allowing for precise tracking of exponential growth.

The goal of this experiment was to quantitatively determine the effect of 4-nitrophenol on the growth of the unicellular green alga *R. subcapitata* by means of a 72-hour freshwater algal growth inhibition test following ISO 8692 [5]. The objective of this study was to calculate algal growth rates and to determine the EC10, EC20, and EC50 values for growth-rate inhibition, including their associated confidence intervals, by comparing exposed cultures with an untreated control.

METHOD

The Water quality - Fresh water algal growth inhibition test (ISO Standard 8692. 2004) were conducted for five different concentrations of 4-nitrophenol. The concentrations were deducted from an initial EC50 value of 30 mg/L (average of literature findings [6][7]) and accordingly reduced or increased by a certain factor (¼, ½, 1, 2, 4) to have a meaningful range of test values.

- Concentration of the stock solution: 2400 mg/L
- Concentration range used: [0, 7.5, 15, 30, 60, 120] mg/L

The principle of the test was to evaluate the toxic effects of the compound on the growth rate and biomass production of freshwater algae when exposed to the different concentrations. To initiate the test, five test solutions were prepared, each with a total volume of 25 mL, yielding an initial algal cell density of $5 \cdot 10^4$cells/mL in each flask. Each solution consisted of a specific volume of stock solution, growth medium, and algal inoculum (See Table 3 and Equation 1 in Appendix).

For the first set of data (under light conditions), the concentrations were tested 3 times each vial and to have a basis for comparison, the test also included five control samples without any addition of the contaminant, making a total of 20 replicates (each with a total volume of 4 mL).

For the second set of data (under dark conditions), 5 blind tubes were prepared, containing 0.4 mL from each test solution and 1.6 mL of acetone and for background correction a further tube with medium only was prepared.

After 72h, approximately, the tubes for biomass measurement were prepared by taking 0.5 mL from each vial and 1.6 mL of acetone, closed cautiously and stored in the dark. With this step, algal growth was stopped.

After 1 more day, the fluorescence of the tubes was measured with the fluorometer. These measurements can be directly used as biomass parameters to define the inhibitory effects of the compound (see values in *Table 4* under Appendix).

Data treatment

The mean growth rate (μ) for each replicate was calculated using the following equation:

$$\mu = \frac{(\ln(N_n) - \ln(N_o)}{t_d}$$

Where:

- t_d is the exposure period in days
- N_o is the initial biomass (fluorescence measurement at the start)
- $N_n$is the final biomass (fluorescence measurement at the end)

The relative inhibition of the growth rate (I) for each concentration was then determined by comparing the exposed cultures to the control:

$$I = (1 - \frac{\mu_i}{\mu_c})$$

Where μ_i is the mean growth rate for concentration i, and μ_c is the mean growth rate for the control.

The concentration-response dataset was analyzed using R software following DTU guidance (see *Figure 1*).

Finally, the three replicates for each concentration were then pooled into one vial, and the pH was measured (see values in Table 5).

RESULTS

The concentration–response data for algal growth inhibition after 72 h has been summarized in *Table 1*.

Table 1:Concentration–response data for algal growth inhibition after 72 h (Derived from Table 4).

Concentration (mg/L)	Mean growth rate, μ [d^{-1}]	Mean inhibition, $I_{\mu i}$
0	1.017	0.0000
7.5	0.6902	0.3213
15	0.3149	0.6903
30	0.2522	0.7520
60	0.1866	0.8165
120	0.0000	1.000

The results show a clear concentration–response relationship for the growth of *R. subcapitata* after 72 hours. The mean growth rate in the control was 1.017 d^{-1}, which indicates normal exponential growth under the test conditions. As the concentration increased, the growth rate decreased steadily, from 0.6902 d^{-1} at 7.5 mg/L to complete inhibition (0 d^{-1}) at 120 mg/L. Correspondingly, the inhibition increased from 32.13% to 100%, demonstrating a strong toxic effect of the test compound at higher concentrations (see the Discussion section for further considerations).

The concentration–response curve for algal growth inhibition after 72h can be observed in *Figure 1* below.

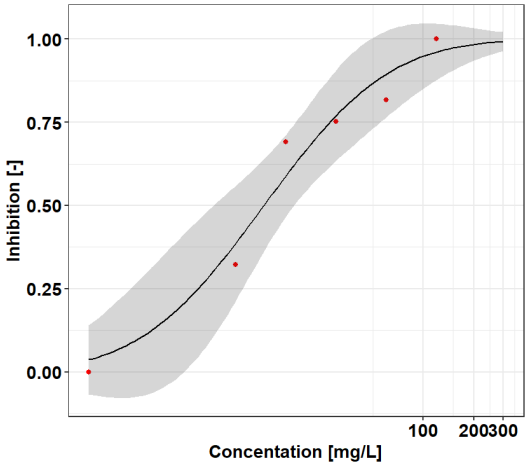


Figure 1: Concentration-inhibition curves after 72h

The concentration–response relationship showed a clear sigmoidal pattern, indicating a typical dose–response behavior for algal growth inhibition. Inhibition was already observed at low concentrations and as the concentration increased, the inhibition effect rose steeply and approached a maximum at higher concentrations. The fitted model showed good agreement with the experimental data, with most observations closely following the predicted curve. The confidence intervals were relatively narrow in the mid-range of concentrations, indicating a high precision of the model in this region. In contrast, slightly wider confidence intervals at the lower and upper concentration ranges reflect increased uncertainty due to fewer data points.

The EC estimated from the 72 h exposure data was gathered in *Table 2*.

Table 2: EC10, EC20 & EC50 values after 72 h.

	Value (mg/L)	SD	Lower (95% CI)	Upper (95% CI)
EC10	1.986	0.9545	-0.6638	4.636
EC20	3.588	1.282	0.02986	7.147
EC50	11.12	1.867	5.940	16.31

The estimated effective concentrations were EC10 = 1.986 mg/L, EC20 = 3.588 mg/L, and EC50 = 11.12 mg/L.

The relatively small standard errors associated with these estimates suggest a high level of precision in the model fitting (see the Discussion section for further considerations).

Other observations

No stimulation (hormesis) was observed at low concentrations, as the growth rates at all tested concentrations remained below that of the control, yielding consistently positive inhibition values. This suggests that the compound exerts inhibitory effects even at the lowest tested concentration, indicating relatively high toxicity and the absence of any stimulatory response. Therefore, a t-test for hormesis was not required.

DISCUSSION

Test Validity and Physical-Chemical Stability

Test validity was confirmed based on growth rates and pH stability. The control growth rate was 1.017 d^{-1}, which securely exceeds the minimum requirement of 0.9 d^{-1}. Additionally, the variations in pH during the test remained well below the acceptable limit of 1.5 units (see Table 5 under Appendix). This indicates that the culture conditions were highly stable, and the observed inhibition can be confidently attributed primarily to the test compound rather than poor algal performance or concentration loss.

Furthermore, the physical-chemical properties of 4-nitrophenol actively supported the integrity of the test

conditions. The compound's extremely low volatility ($H<10^{-5}$atm-m³/mol) ensured that the actual concentrations did not deviate significantly from the nominal concentrations due to evaporation, and it also ensures that the concentration stayed stable over the 72h allowing good reproducibility of the test [3]. This supports the reliability of the EC values obtained from the study.

Moreover, the moderate water solubility of 16 g/L allowed the compound to be fully dissolved into the solution, ensuring that nominal concentration was the actual concentration [3]. This improved the accuracy and consistency of the exposure to the *R. subcapitata;* additionally, the absence of precipitates restricted the possibility of light scattering and false fluorescence readings. The final readings may nevertheless be affected by minor factors such as sorption and photodegradation. Considering that the compound has a pKa value of 7.15, 4-nitrophenol exhibits intermediate dissociation behavior, with both neutral and ionized forms present This results in moderate sorption properties, meaning that some interaction with vial walls is possible; but most of the compound remains dissolved and bioavailable [8]. Lastly, photodegradation must be considered since the test vials were exposed to strong light conditions which could have led to a reduction in the actual concentration over the 72h period and the formation of degradation products with potentially different toxicity [9].

Concentration-Response Analysis and Limitations

To improve the reliability of the results, background correction was applied using the measured fluorescence from the blind vial containing only the culture medium. This correction is essential to eliminate non-biological signals, ensuring that fluorescence measurements at day 0 and after 72 h accurately represent the initial biomass and true growth of the test organism, respectively. Variations in the measured values may arise from instrument noise or from the presence of the tested solution itself. Background correction was performed by subtracting the blind measurement from each recorded fluorescence value. As the background signal was relatively small, this correction resulted in only minor changes (the last 2 significant figures) to the overall values.

Overall, the experiment was conducted without major complications, resulting in a clear sigmoidal concentration-response curve. Increasing concentrations of 4-nitrophenol resulted in progressively higher inhibitory effects on *R. subcapitata,* starting with a 32.13% inhibition at the lowest concentration (7.5 mg/L) and reaching complete 100% inhibition at 120 mg/L. This mathematically valid curve allowed for the successful determination of the EC50 value (11.12 mg/L) with small confidence intervals, confirming that the reference expected EC50 of 30 mg/L was an adequate starting point for the test design.

A few limitations regarding the low-effect estimates must also be noted. The lowest tested concentration (7.5 mg/L) already induced over 32% inhibition, leaving a gap in data points between 0% and 32%. Because the selected range did not fully capture the gradual onset of inhibition, the lower part of the curve was less constrained. This uncertainty is reflected in the confidence interval for the EC10, which resulted in a negative lower bound. Therefore, while the EC50 estimate is highly robust, the EC10 value is less precise and would have benefited from a more finely spaced concentration series below 7.5 mg/L.

Comparison with Literature EC-values for 4-Nitrophenol

According to the EPA Hazard Characterization and ANZ Guidelines data collected in previous sessions, the acute toxicity of 4-nitrophenol across different trophic levels is relatively high. Regarding the 72-h/96-h EC50 for aquatic plants (e.g., *Desmodesmus subspicatus*), literature values were found to be > 32 mg/L [10].

Although this may initially appear inconsistent with our experimentally determined 72-h EC50 of 11.12 mg/L, it is important to note that EC50 values are not a fixed property of a chemical. They depend heavily on exposure duration and the specific sensitivity of the organism. The literature finding is a range-based estimate generally applied to aquatic plants, and substantial biological differences between species naturally result in different toxicological endpoints. This observation does not fully follow the general expectation that shorter exposure

times typically result in higher EC50 values and this discrepancy may be attributed to differences in species sensitivity as previously mentioned, experimental conditions, or variability in the test system.

CONCLUSION

The 72-hour freshwater algal growth inhibition test successfully quantified the toxicity of 4-nitrophenol on the green microalga *R. subcapitata*. The experiment fulfilled all standardized validity criteria, maintaining a stable environment (ΔpH < 1.5) and demonstrating a healthy control growth rate of 1.017 d^{-1} [1].

The results revealed a clear, concentration-dependent sigmoidal inhibition of algal growth, with no stimulatory effects (hormesis) observed at lower doses [11]. At the maximum tested concentration of 120 mg/L, full inhibition was observed, indicated by negative fluorescence values and a growth rate of 0 d^{-1}.

The reliable model fit allowed for the precise determination of key ecotoxicological endpoints: an EC10 of 1.986 mg/L, an EC20 of 3.588 mg/L, and an EC50 of 11.12 mg/L.

While the selected concentration range was highly effective for deriving a robust EC50, the wider confidence intervals at the lower boundary suggest that future assessments could benefit from a narrower concentration gradient below 7.5 mg/L to refine the EC10 estimate. Ultimately, the derived EC values indicate that 4-nitrophenol is highly toxic to primary producers, providing essential quantitative data for environmental risk assessments and the establishment of future regulatory thresholds in aquatic ecosystems.

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APPENDIX

A1. Tables

Table 3. Preparation quantities for the different test solution.

Target Conc. (mg/L)	Stock Volume (mL)	Stock µL	Inoculum (mL)	Medium Volume (mL)	Final Volume (mL)
0	0	0	1.25	23.75	25
7.5	0.078125	78.125	1.25	23.67	25
15	0.15625	156.25	1.25	23.59	25
30	0.3125	312.5	1.25	23.44	25
60	0.625	625	1.25	23.13	25
120	1.25	1250	1.25	22.5	25

Table 4. Raw data retrieved from the tests performed and calculated growth rate, average and inhibition.

Concentration mg/L	Replicate	Day 0 Biomass reading, N_o	72h Biomass reading, N_n	Growth rate (µ)	Average µ	Inhibition, Iµi
0	A	36.21	942.7	1.0864	1.0168	-0.0673
0	B	36.21	634.1	0.9543		0.0627
0	C	36.21	758.9	1.0142		0.0038
0	D	36.21	687.8	0.9814		0.0360
0	E	36.21	839.3	1.0477		-0.0292
7.5	A	24.12	186.8	0.6822	0.6901	0.3307
7.5	B	24.12	173.2	0.6570		0.3554
7.5	C	24.12	216.2	0.7310		0.2828
15	A	29.22	52.32	0.1942	0.3149	0.8097
15	B	29.22	94.91	0.3927		0.6149
15	C	29.22	85.49	0.3578		0.6491
30	A	23.88	58.79	0.3003	0.2522	0.7058
30	B	23.88	67.11	0.3444		0.6625
30	C	23.88	33.39	0.1117		0.8907
60	A	22.84	37.24	0.1630	0.1866	0.8405
60	B	22.84	45.16	0.2272		0.7775
60	C	22.84	38.00	0.1697		0.8339

120	A	13.68	-1.759	0.0000	0	1.0000
120	B	13.68	-1.843	0.0000		1.0000
120	C	13.68	-2.318	0.0000		1.0000

Table 5. pH values measured after 72 h for each test concentration.

Concentration (mg/L)	pH values
0	6.40
7.5	7.35
15	7.48
30	7.62
60	7.67
120	7.60

A2. Equations and calculations

Equation 1.

The volume of inoculum needed into the flask was deducted from: $C_{pre} * V_{pre} = C_{flask} * V_{flask}$

Where:

- V_{pre} is the volume of the pre-culture [mL]
- C_{pre} is the concentration of algal cells in pre-culture = 10^6 cells/mL
- C_{flask} is the concentration of algal cells in the 25 mL flask = $5 * 10^4$ cells/mL
- V_{flask} is the volume of the flask = 25 mL